Giovanni De Felice

Post-doc

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Languages

- Italian - Native
- English - Professional
- French - Basic

Short Bio

Postdoctoral researcher at the Università della Svizzera Italiana (CH). My research focuses on:

- **Interpretability-by-design:** model design, theory, and code.
- **Deep Learning for Time Series:** forecasting, classification, representation, spatio-temporal data, virtual sensing.

Experience

2025 - Ong. **Postdoctoral researcher** (Prof. S. Santini)
Università della Svizzera italiana, Lugano, (Switzerland)
- Concept-based interpretable models.
- Machine Learning and interpretability for time series.

2024 **Research visit** (Prof. C. Alippi)
Swiss AI lab IDSIA & USI, Lugano (Switzerland)
- Graph deep learning for time series.

2020 - 2024 **Ph.D. candidate** in Computer Science
University of Liverpool, Liverpool (UK)
- Machine learning for time series & spatio-temporal data.
- Applications in the natural sciences.

2020 **DOE-INFN Summer Intern** (Dr. R. H. Bernstein)
Fermilab, Batavia, IL (USA)
- Antiproton background in the Mu2e beamline.

2018 - 2020 **Master of Science (M.Sc.)** in Particle Physics
University of Pisa, Pisa (Italy)
Final grade: 110/110 cum Laude, GPA: 29.5/30
- Experimental particle physics.
- Statistical data analysis.

Selected publications

- **Interpretability in Deep Time Series Models Demands Semantic Alignment**
  G. De Felice, R. D'Elia, A. Termine, P. Barbiero, G. Marra, S. Santini.
  arXiv preprint arXiv:2602.02239. 2026
- **Causally Reliable Concept Bottleneck Models**
  G. De Felice, A. Casanova, F. De Santis, S. Santini, J. Schneider, P. Barbiero, A. Termine.
  Advances in Neural Information Processing Systems (NeurIPS). 2025
- **Graph-based Virtual Sensing from Sparse and Partial Multivariate Observations**
  G. De Felice, A. Cini, D. Zambon, V. Gusev, C. Alippi.
  International Conference on Learning Representations (ICLR). 2024.
- **Time Series Kernels based on Nonlinear Vector AutoRegressive Delay Embeddings**
  G. De Felice, J. Y. Goulermas, and V. Gusev.
  Advances in Neural Information Processing Systems (NeurIPS). 2023.

Talks & dissemination

- **Invited talk** at **ETH Zurich** (Mar 2026,).
  "PyTorch Concepts: Towards Standard Interpretable Machine Learning".
- **Guest lecture** at **Università della Svizzera Italiana** (Mar 2026,).
  "Standard Interpretable Machine Learning", in "Advanced Topics in Machine Learning" (Prof. Cesare Alippi).
- **Invited talk** at **KU Leuven** (Feb 2026,).
  "PyTorch Concepts: interpretable and causally transparent deep learning models."
- **Invited talk** at **University of Cambridge** (Feb 2026,).
  "Towards Causally Reliable Concept-based Models".
- **Invited talk** at *'Temporal Graph Reading Group'* (Mar 2024, Virtual).
  Presentation of the paper: "Graph-based Virtual Sensing from Sparse and Partial Multivariate Observations".
- **Invited talk** at **Beckers group** (Nov 2023,).
  Invited to present my work to the Executive Management team of the Beckers Group.
- **Invited talks** at two *Mu2e Collaboration Meetings* (Oct 2020, and Jun 2020, Virtual).
- **Poster presentations** at PGR Workshop (Univ. of Liverpool), NeurIPS 2022 (New Orleans), NeurIPS 2023 (New Orleans), ICLR 2024 (Vienna), NeurIPS 2025 (San Diego).

Program committee member

- **Reviewer** for *International Conference on Neural Information Processing Systems* (NeurIPS)
- **Reviewer** for *International Conference on Machine Learning* (ICML)
- **Reviewer** for *International Conference on Learning Representations* (ICLR)
- **Reviewer** for *IEEE Transactions on Neural Networks and Learning Systems* (TNNLS) journal

Industrial projects

2020 - 2024 **Weathering Predictions of Paint Formulations**, in partnership with **Beckers Group**.
Beckers Group / University of Liverpool
- Predict long-term performances in untested locations from climatic data.
- Extract formulatory information from data.

Code

2025 - Ong. Reference author of **PyTorch Concepts**

- Library to easily implement concept-based interpretable models.
- `https://github.com/pyc-team/pytorch_concepts/tree/master`

Other interests

Music and Classical Piano: I started playing the piano when I was 8 yo, reached a high conservatory level and it still constitutes a central part of my life.

- Classical Piano Degree Admission ($\sim 8^{th}/10$ year exam)
  Higher institute of musical studies Pietro Mascagni, Livorno, Italy. Grade: 8/10 **(2nd place)** (2015)
- Mid-term exam ($5^{th}/10$ year exam)
  Conservatory Alfredo Casella, l'Aquila, Italy. Grade: 9/10 (2013)
- Solfeggio and theory of music
  Conservatory Nino Rota, Monopoli, Bari, Italy (2011)
- GRADE 1 exam
  The Associated Board of the Royal Schools of Music, Varese, Italy. Grade: pass with distinction (2007)
- Other: Alto Saxophone (2022-Ong.), keyboard live concerts in Italy and France (2013-2015), pianist in a theater-dance spectacle (2014), orchestra and chamber music (2010-2015).

All publications

Machine Learning:

- **A foundation model for electrodermal activity data**
  L. Alchieri, M. Garzon, L. Alecci, F. B. De Bona, M. Gjoreski, **G. De Felice**, S. Santini
  arXiv preprint arXiv:2603.16878. 2026
- **Interpretability in Deep Time Series Models Demands Semantic Alignment**
  G. De Felice, R. D'Elia, A. Termine, P. Barbiero, G. Marra, S. Santini.
  arXiv preprint arXiv:2602.02239. 2026
- **Mixture of Concept Bottleneck Experts**
  F. De Santis, G. Ciravegna, **G. De Felice**, A. Casanova, F. Giannini, M. Diligenti, M. Zarlenga, P. Barbiero, J. Schneider, D. Giordano
  arXiv preprint arXiv:2602.02886. 2026
- **Federated Concept-Based Models: Interpretable models with distributed supervision**
  D. Fenoglio, A. Casanova, F. De Santis, M. Li, G. Dominici, J. Schneider, M. Gjoreski, M. Langheinrich, P. Barbiero, **G. De Felice**
  arXiv preprint arXiv:2602.04093. 2026
- **Causally Reliable Concept Bottleneck Models**
  G. De Felice, A. Casanova, F. De Santis, S. Santini, J. Schneider, P. Barbiero, A. Termine.
  Advances in Neural Information Processing Systems (NeurIPS). 2025
- **On the Regularization of Learnable Embeddings for Time Series Processing**
  L. Butera, **G. De Felice**, A. Cini, C. Alippi.
  Transactions on Machine Learning Research (TMLR). 2025.
- **Graph-based Virtual Sensing from Sparse and Partial Multivariate Observations**
  G. De Felice, A. Cini, D. Zambon, V. Gusev, C. Alippi.
  International Conference on Learning Representations (ICLR). 2024.
- **Time Series Kernels based on Nonlinear Vector AutoRegressive Delay Embeddings**
  G. De Felice, J. Y. Goulermas, and V. Gusev.
  Advances in Neural Information Processing Systems (NeurIPS). 2023.

- **Spatio-Temporal Weathering Predictions in the Sparse Data Regime with Gaussian Processes**
  G. De Felice, V. Gusev, J. Y. Goulermas, M. Gaultois, M. Rosseinsky, C. Gauvin,
  AI for Science: Progress and Promises (NeurIPS Workshop). 2022.

Physics:

- **Mu2e Run I Sensitivity Projections for the Neutrinoless $\mu^- \to e^-$ Conversion Search in Aluminum**
  Mu2e Collaboration,
  Universe, 9(1), p.54. 2023.

- **An updated estimate of the Mu2e experiment sensitivity**
  G. De Felice,
  Master's degree thesis. 2020.

Materials Informatics:

- **Enhancing Extrapolation in Materials Science through Contrastive Learning of Chemical Compositions**
  O. Federico, **G. De Felice**, R. Savani, V. Gusev, and M. Rosseinsky.
  AI for Accelerated Materials Design (NeurIPS Workshop). **Spotlight**. 2023.

- **Not as simple as we thought: a rigorous examination of data aggregation in materials informatics**
  O. Federico, **G. De Felice**, V. Gusev, and T. Sparks.
  Digital Discovery, 3 (2), p.337-346. 2023.